HHN Business Intelligence 20261105

Convolution applications:

- Computer vision
- AI for graphics

1. A BI system analyzes a small greyscale icon patch with a local contrast filter.

a) Given

$$X = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 4 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

compute the top-left output entry $Y_{1,1}$.

⇒ To get $Y_{1,1}$ in a valid convolution, we take the top-left 2×2 submatrix and apply filter K.

$$X_{1:2,1:2} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad K = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$Y_{1,1} = 1 \cdot 1 + 2 \cdot (-1) + 0 \cdot (-1) + 3 \cdot (1)$$

$$= 1 - 2 + 0 + 3$$

$$\boxed{Y_{1,1} = 2} \quad ✓$$

b) Compute the full valid convolution output matrix Y

⇒ $Y_{1,1} = 2$

⇒ $Y_{1,2} = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$$= 2 \cdot 1 + 0 \cdot (-1) + 3 \cdot (-1) + 1 \cdot 1$$

$$\boxed{Y_{1,2} = 0} \quad ✓$$

$\Rightarrow Y_{2,1} = \begin{vmatrix} 0 & 3 \\ 2 & 1 \end{vmatrix} \odot \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix}$

$= 0\cdot(1) + 3\cdot(-1) + 2(-1) + 1\cdot 1$ **$= -4$** ✓

$\Rightarrow Y_{2,2} = \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} \odot \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix}$

$= 3\cdot 1 + 1\cdot(-1) + 1\cdot(-1) + 4\cdot 1$ **$= 5$** ✓

$$Y = \begin{vmatrix} Y_{1,1} & Y_{1,2} \\ Y_{2,1} & Y_{2,2} \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ 4 & 5 \end{vmatrix}$$ ✓

c) Determine the position of the largest response and explain what this says about the local pattern in the input.

$\Rightarrow$ The position of the largest response is at **$Y_{2,2} = 5$**

$\Rightarrow$ Since we have a diagonal contrast filter, it gives a high positive value when:

- The values on the main diagonal are high (3 and 4).
- The values on the anti-diagonal are low.

$$Y_{2,2} \Rightarrow \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix}$$

2 A dashboard image is processed with a vertial edge detector.

a) Given

$$X = \begin{vmatrix} 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \end{vmatrix}, \quad K = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{vmatrix},$$

compute $Y_{1,1}$, $Y_{1,2}$.

$\Rightarrow$ $Y_{1,1}$ corresponds to the top-left submatrix

$$X_{1:3,1:3} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}, \quad K = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{vmatrix}$$

$\Rightarrow$ $Y_{1,1} = 1 \cdot 1 + 1 \cdot 0 + 1 \cdot (-1) = 0$ $\leftarrow$ every row

$$\boxed{Y_{1,1} = 0} \checkmark$$

$\Rightarrow$ We then slide the kernel over one column to the right to get $Y_{1,2}$.

$\Rightarrow$ $$X_{1:3,2:4} = \begin{vmatrix} 1 & 1 & 4 \\ 1 & 1 & 4 \\ 1 & 1 & 4 \end{vmatrix}, \quad K = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{vmatrix}$$

$\Rightarrow$ Each row: $1 \cdot 1 + 1 \cdot 0 + 4(-1) = -3$

$$\boxed{Y_{1,2} = -3 - 3 - 3 = -9} \checkmark$$

b) Compute the full valid feature map Y.

$\Rightarrow$ Since we are sliding a 3×3 kernel over a 4×4 input matrix, the output dimensions for the valid convolution feature map are:

$$\text{output_d} = (d_x - d_k + 1)$$
$$= (4 - 3 + 1) = 2 \checkmark$$

$\Rightarrow$ Y is a 2×2 matrix.

$\Rightarrow$ $Y_{1,1} = 0$ $\checkmark$

$\Rightarrow$ $Y_{1,2} = -9$ $\checkmark$

$\Rightarrow$ $Y_{2,1} \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$ $\Rightarrow$ This is identical to the calc. for $Y_{1,1}$

$$\boxed{Y_{2,1} = 0} \checkmark$$

$\Rightarrow Y_{2,2} \Rightarrow \begin{vmatrix} 1 & 1 & 4 \\ 1 & 1 & 4 \\ 1 & 1 & 4 \end{vmatrix} \Rightarrow \boxed{Y_{2,2} = -9}$ Identical to calc. for $Y_{1,2}$!

$$\boxed{Y = \begin{vmatrix} 0 & -9 \\ 0 & -9 \end{vmatrix}} \checkmark$$

c) Explain what the sign of the entries means and state where the vertical edge is located.

⇒ The sign indicates the direction of intensity change (gradient) relative to the filter's location.

⇒ In the filter K, the right column has negative weights (-1 entries) and the left column has positive weights (+1).

⇒ A negative result means that the pixels on the right side of the image are much brighter (more intense) than those on the left.

⇒ Location of vertical edge: the edge is located in the transition from column 3 to column 4.

Reading ⇒ Google: YOLO OpenCV

5 A greyscale patch is processed with a Laplacian filter using same padding.

a) Given

$$X = \begin{vmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{vmatrix}, \qquad K = \begin{vmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{vmatrix},$$

write the zero-padded matrix $\tilde{X}$

$\Rightarrow$ To achieve same-padding for the 3x3 input X, with a 3x3 filter, we use a border of 1 pixel of zeros on all sides

$$\tilde{X} = \begin{vmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{vmatrix}$$

b) Compute $Y_{1,1}$, $Y_{2,2}$, $Y_{3,3}$

$\Rightarrow$ $Y_{1,1} = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{vmatrix} \odot \begin{vmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{vmatrix}$ $= 0\cdot 0 + 0\cdot(-1) + 0\cdot 0 + 0\cdot(-1) + 0\cdot 4 + 1\cdot(-1) + 0\cdot 0 + 1\cdot(-1) + 2\cdot(0)$

$$Y_{1,1} = -1 - 2 + 1 = -2$$ ✓

$\Rightarrow$ $Y_{2,2} = \begin{vmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{vmatrix} \odot \begin{vmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{vmatrix}$ $= 0\cdot 0 + 1\cdot(-1) + 0\cdot 0 + 1\cdot(-1) + 2\cdot 4 + 1\cdot(-1) + 0\cdot 0 + 1\cdot(-1) + 0\cdot 0$

$$Y_{2,2} = -1 + 6 - 1 = 4$$ ✓

$\Rightarrow$ $Y_{3,3} = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} \odot \begin{vmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{vmatrix}$ $= 2\cdot 0 + 1\cdot(-1) + 0\cdot 0 + 1\cdot(-1) + 0\cdot 4 + 0\cdot(-1) + 0\cdot 0 + 0\cdot(-1) + 0\cdot 0$

$$Y_{3,3} = 0 - 1 - 1 + 0 = -2$$ ✓

c) Compute the full same-padded output Y, and identify the position with the strongest positive response.

$\Rightarrow$ We already have $Y_{1,1}$, $Y_{2,2}$, $Y_{3,3}$

$\Rightarrow$ We will use the property of the Laplacian filter that is the sum-of-neighbors comparison

$$K = \begin{vmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{vmatrix}$$

$\Rightarrow$ The center (4) is the multiplier for the "target" pixel.

$\Rightarrow$ The neighbors of the "target" pixel are subtracted (-1).

$\Rightarrow$ $Y_{1,2}$: $\begin{vmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{vmatrix}$, Target: 1, neighbors: (0, 0, 0, 2)

$$Y_{1,2} = 1 \cdot 4 - (0 + 0 + 0 + 2) = 2$$ ✓

$\Rightarrow$ $Y_{1,3}$: $\begin{vmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{vmatrix}$, Target: 0, neighbors: (0, 1, 0, 1)

$$Y_{1,3} = 0 \cdot 4 - (0 + 1 + 0 + 1) = -2$$

$\Rightarrow$ $Y_{2,1}$: $\begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix}$, Target: 1, neighbors: (0, 0, 2, 0)

$$Y_{2,1} = 1 \cdot 4 - (0 + 0 + 2 + 0) = 2$$

$\Rightarrow$ $Y_{2,3}$: $\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}$, Target: 1, neighbors: (0, 2, 0, 0)

$$Y_{2,3} = 1 \cdot 4 - (0 + 2 + 0 + 0) = 2$$

$\Rightarrow$ $Y_{3,1}$: $\begin{vmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{vmatrix}$, Target: 0, neighbors: (1, 0, 1, 0)

$$Y_{3,1} = 0 \cdot 4 - (1 + 0 + 1 + 0) = -2$$

$\Rightarrow Y_{3,2}$: $\begin{vmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{vmatrix}$, Target: 1, neighbors: (2, 0, 0, 0)

$$Y_{3,2} = 1 \cdot 4 - (2 + 0 + 0 + 0) = 2$$

$$Y = \begin{vmatrix} -2 & 2 & -2 \\ 2 & 4 & 2 \\ -2 & 2 & -2 \end{vmatrix}$$ ✓

$\Rightarrow$ The strongest positive response is 4, at position $Y_{2,2}$!

$\Rightarrow$ The high positive value indicates a spike or bright spot.

$\Rightarrow$ The positive neighbors (2s) are ridges around the spot

$\Rightarrow$ The negative entries at the corners (-2s) represent steep drop-off in intensity.

10 An RGB patch is processed with one $2 \times 2 \times 3$ filter.

a) The RGB channels are

$$R = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \quad G = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \quad B = \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix}$$

the filter channels:

$$K_R = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \quad K_G = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \quad K_B = \begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix}$$

Compute the contribution of the R-channel to the output scalar.

$\Rightarrow$ Since the filter is 2×2 and the input is 2×2, we only have one filter position.

$\Rightarrow$ No sliding.

$$R \Rightarrow \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \odot \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1 + 2 \cdot 0 + 0 \cdot 0 + 1 \cdot 1$$

$$= 1 + 0 + 0 + 1$$

Contr_R

$= 2$ ✓

b) Compute the G and B channel contributions.

$G \Rightarrow \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \odot \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0$

Contr_G $= 2$ ✓

$B \Rightarrow \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} \odot \begin{vmatrix} +1 & 0 \\ 0 & -1 \end{vmatrix} = 2 \cdot (-1) + 0 \cdot 0 + 1 \cdot 0 + 1 \cdot (-1)$

Contr_B $= -3$ ✓

c) Compute the final output scalar and interpret whether the patch is more red-green dominant or more blue dominant with respect to this filter.

$\Rightarrow$ To get the final output scalar, we sum the contributions of all 3 channels:

$$Y = \text{Contr_R} + \text{Contr_G} + \text{Contr_B}$$

$$Y = 2 + 2 - 3 = 1$$ ✓

$\Rightarrow$ The patch is clearly red-green dominant, because the contributions from R and G are positive, while that of B is negative.

11 A BI team segments four customers with $K = 2$.

a) Given the points

$A(1,1)$, $B(1,2)$, $C(5,4)$, $D(6,5)$,

and initial centroids

$m_1^{(0)} = (0,0)$, $m_2^{(0)} = (6,4)$,

compute the distances from each point to both centroids, and assign each point to a cluster.

Rules:
- Use squared Euclidean distances
- Break ties for smaller cluster idx
- Update centroids at least once

Point	$d^2(x, m_1)$	$d^2(x, m_2)$	Winner
$A(1,1)$	$(1-0)^2 + (1-0)^2 = 2$	$(1-6)^2 + (1-4)^2 = 34$	C_1
$B(1,2)$	$(1-0)^2 + (2-0)^2 = 5$	$(1-6)^2 + (2-4)^2 = 29$	C_1
$C(5,4)$	$(5-0)^2 + (4-0)^2 = 41$	$(5-6)^2 + (4-4)^2 = 1$	C_2
$D(6,5)$	$(6-0)^2 + (5-0)^2 = 61$	$(6-6)^2 + (5-4)^2 = 1$	C_2

Assignments $\Rightarrow$ C_1: $\{A, B\}$
C_2: $\{C, D\}$

b) Compute the updated centroids

$m_1^{(1)}$, $m_2^{(1)}$.

$\Rightarrow$ To get the new centroids, we take the arithmetic mean of all points in a cluster:

$$\frac{1}{|C_k|} \sum_{x \in C_k} x$$

$C_1 \Rightarrow A(1,1)$, $B(1,2)$ $\Rightarrow$ $x = \frac{1+1}{2} = 1$ ✓

$y = \frac{1+2}{2} = 1.5$ ✓

$\Rightarrow m_1^{(1)} = (1, 1.5)$

$C_2 \Rightarrow C(5,4)$, $D(6,5)$ $\Rightarrow$ $x = \frac{5+6}{2} = 5.5$

$y = \frac{4+5}{2} = 4.5$

$\Rightarrow m_2^{(1)} = (5.5, 4.5)$



c) Perform one more assignment step with the updated centroids and decide whether the algorithm has converged.

Point	$d^2(x, m_1)$	$d^2(x, m_2)$	Winner
A(1, 1)	$(1-1)^2 + (1-1.5)^2 = 0.25$	$(1-5.5)^2 + (1-4.5)^2 = 32.5$	C_1
B(1, 2)	$(1-1)^2 + (2-1.5)^2 = 0.25$	$(1-5.5)^2 + (2-4.5)^2 = 26.5$	C_1
C(5, 4)	$(5-1)^2 + (4-1.5)^2 = 22.25$	$(5-5.5)^2 + (4-4.5)^2 = 0.5$	C_2
D(6, 5)	$(6-1)^2 + (5-1.5)^2 = 37.25$	$(6-5.5)^2 + (5-4.5)^2 = 0.5$	C_2

Assignments $\Rightarrow$ C_1: {A, B}
C_2: {C, D}

$\Rightarrow$ Convergence !